University for the Creative Arts

BERLIN SCHOOL OF BUSINESS & INNOVATION

Essay / Assignment Title: **Multi-Touch Attribution Story**

Programme Title: **MSc Data Analytics**

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Year: **2026**

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Date: 14 – February – 2026

1. PROJECT GOAL

In online marketing, a user interacts with more than one channel before a conversion takes place. When credit is given only to the first touch or the last touch, the impact may be assigned to the wrong channel. This happens because the entire journey is not taken into account.

This project uses a multi-touch attribution dataset to study the relation between marketing channels, touchpoint sequences, and conversion probability. The aim is to build Tableau dashboards and a story. These will help business users compare channels by their role in the journey and find which paths are linked with higher conversion rates.

Main research question:

Which marketing channels and touchpoint sequences are linked with higher conversion probability?

2. LITERATURE REVIEW

Marketing attribution looks at how conversion credit should be shared across many touchpoints. Many firms still use rule-based methods like last-touch or first-touch because they are simple to apply. But these rules do not use the full journey, so they can undervalue early or assisting channels (Berman, 2018).

Studies have developed data-driven multi-touch attribution (MTA) models. These models use the actual journey paths that users follow to estimate how each channel links to conversion. The models apply probability-based, regression-based, or structural methods. They divide credit across all the exposures in the journey. This is different from giving full credit to only one interaction (Abhishek, Fader and Hosanagar, 2017; Xu et al., 2016).

This approach supports the idea that attribution should consider the complete journey, not just a single touchpoint. A key issue is that path data is observational. Users choose channels based on their own intent, and firms may target users with different intent levels, which creates endogeneity. Studies show that simple attribution models can overstate channel value if they do not control for selection bias and cross-channel effects (Ghose and Todri, 2016; Abhishek, Fader and Hosanagar, 2017). So, a high conversion rate on a path does not always mean the path caused the conversion.

Another line of work treats journeys as sequences. Markov chain and graph-based methods model channels as states and estimate a channel's contribution by simulating what happens when that channel is removed from the transition structure (Anderl et al., 2016; Xu et al., 2016). This helps analyse ordering effects, channel interactions, and how results differ for short versus long journeys.

Research also studies how attribution rules shape incentives. Last-touch attribution can push publishers to compete for the final click and can reduce advertiser value. Game-theory methods like Shapley value sharing can allocate credit more fairly across channels (Berman, 2018).

Based on this literature, this project uses Tableau to provide descriptive MTA evidence, such as first-touch and last-touch conversion rates, journey length patterns, and top sequences. The dashboards are built to support decisions and generate test ideas, while avoiding causal claims without controlled experiments.

3. DATASET OVERVIEW AND PREPROCESSING

Data Source and File

The dataset comes from Kaggle. The file name is multi_touch_attribution_data.csv. Each row is one touchpoint event. It records one marketing exposure or interaction for a user and includes a timestamp.

Data Integrity

In this project, the dataset is taken as reliable. It is from Kaggle, a site used often for studies and business work. The analysis assumes that all records are present for the given time period and that the main fields are entered correctly. These fields are User ID, Timestamp, Channel, and Conversion. The findings therefore show patterns that are present in this dataset. No other checks were done except the cleaning and changes needed to use the data in Tableau.

Main fields used in this project

- User ID: used to group events into one user journey
- Timestamp: used to order touchpoints within each user journey
- Channel: the marketing channel for the touchpoint
- Campaign: used for segmentation and comparison
- Conversion (Yes/No): outcome flag available at event level

Record counts

- Touchpoint rows: 10,000
- Unique users: 2,847
- Unique channels: 6
- Unique campaigns: 6

Preprocessing summary

This project uses two related tables in Tableau. Data Source A (raw touchpoints): multi_touch_attribution_data.csv. Used for basic KPIs, first-touch view, last-touch view, and journey length analysis. Data Source B (path summary): path_stats.csv. Used for top sequences and path volume versus conversion analysis.

Why preprocessing was needed

The raw file stores touchpoints at the event level. For sequence analysis, Tableau works better when we also have a path-level table. In this table, each row is one complete channel sequence, along with user count and conversion rate for that sequence.

Steps performed (Python)

A Python script was used to create the path-level table. It follows these steps:

- Load the CSV and parse Timestamp as datetime.
- Sort rows by User ID and Timestamp to get the touch order.
- Convert Conversion (Yes/No) into a numeric flag (1/0).
- Create a user-level journey table with:
  - Journey path as a single string
  - Journey length (number of touchpoints)
  - User converted flag (max conversion within the user's events)
  - First touch channel and last touch channel
- Aggregate journeys into path-level statistics (path_stats.csv) with:
  - Journey_Path
  - Users_in_Path
  - Conversion_Rate (mean of user-level conversion for that path)

Python Code

```python
import pandas as pd
df = pd.read_csv("multi_touch_attribution_data.csv")

# Parse timestamp safely
df["Timestamp"] = pd.to_datetime(df["Timestamp"], errors="coerce", dayfirst=True)

# Sort by user + time
df = df.sort_values(["User ID", "Timestamp"])

# Conversion to numeric 0/1
df["ConversionNumeric"] = (df["Conversion"].astype(str).str.strip().str.lower() == "yes").astype(int)

# Build user-level journey dataset
user_journeys = df.groupby("User ID").agg(
    Journey_Path=("Channel", lambda x: " → ".join(x.astype(str))),
    Journey_Length=("Channel", "count"),
    User_Converted=("ConversionNumeric", "max"),
    First_Touch_Channel=("Channel", "first"),
    Last_Touch_Channel=("Channel", "last"),
)
user_journeys = user_journeys.reset_index()

# Add user count per path (useful in Tableau)
path_stats = user_journeys.groupby("Journey_Path").agg(
    Users_in_Path=("User ID", "count"),
    Conversion_Rate=("User_Converted", "mean")
).reset_index()

# Save file (only path_stats.csv needed)
path_stats.to_csv("path_stats.csv", index=False)
```

Outputs used in Tableau

path_stats.csv was used as a separate Tableau data source for sequence analysis.

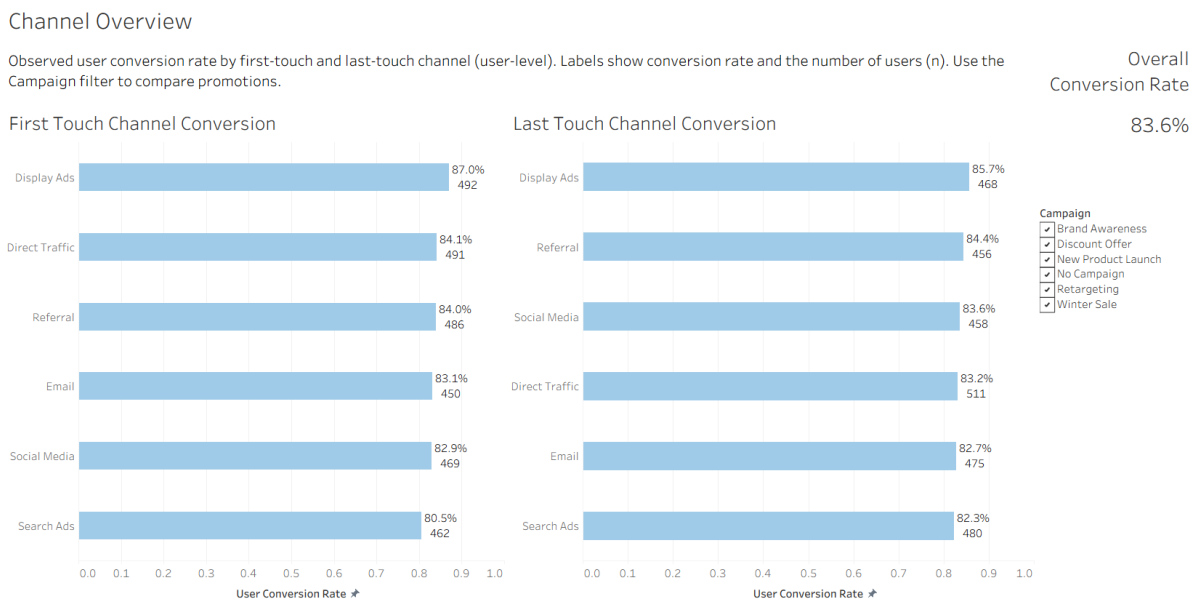
Data quality notes

Timestamp parsing used errors="coerce". If any timestamp cannot be parsed, it becomes null (NaT). This can affect ordering and first/last-touch logic for those users.

If a user has multiple events with the exact same first or last timestamp, more than one row may be marked as first/last when using Tableau LOD logic. This is a tie case and should be treated as descriptive only.

4. KEY VISUALIZATION INSIGHTS AND ANALYSIS

4.1.Baseline context (overall conversion)



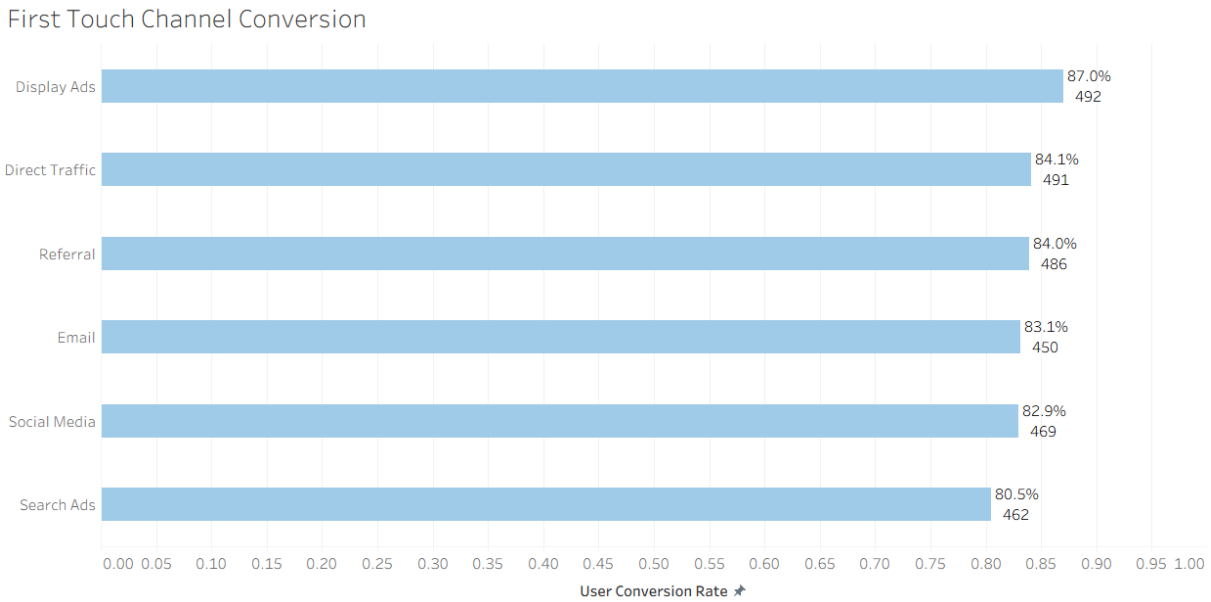
The overall user conversion rate is 83.6%. A user is counted as converted if they have at least one "Yes" conversion event in their full journey.

This baseline is the reference for the full dashboard. It shows that the dataset has a high share of converted users, so channel-to-channel differences are expected to be moderate.

4.2.First-touch channel patterns (entry channels)

The first-touch chart shows conversion rate based on the first channel a user interacted with. Conversion rates by entry channel are close to each other, which means the first channel alone does not create a large performance gap in this dataset.

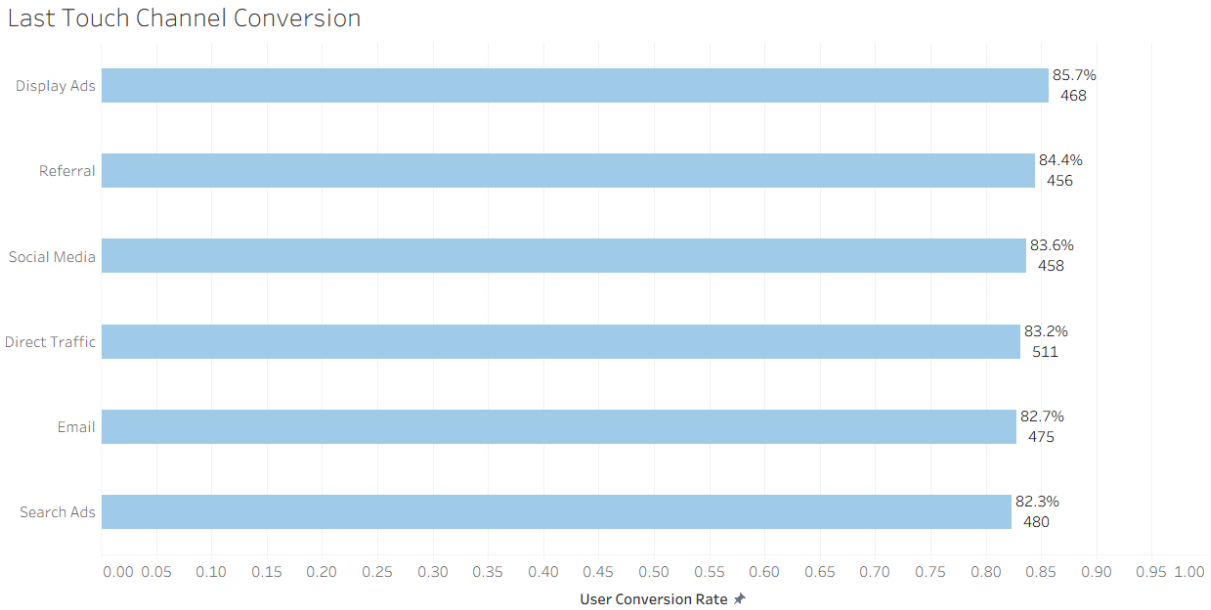
Display Ads has the highest first-touch conversion rate (87.0%, n≈492), while Search Ads is the lowest (80.5%, n≈462). Other channels sit in a narrow band around ~83–84%. Because each channel has a similar number of users, the comparison is based on meaningful volume and supports stable interpretation.



4.3.Last-touch channel patterns (closing channels)

The last-touch chart shows conversion rate based on the last channel a user interacted with. Similar to the first-touch view, the conversion rates show a narrow spread across channels.

Display Ads is highest as last touch (85.7%, n≈468), and Search Ads is lowest (82.3%, n≈480), with other channels close in between. The main point is that single-position views (only first or only last) do not separate channel performance strongly here.

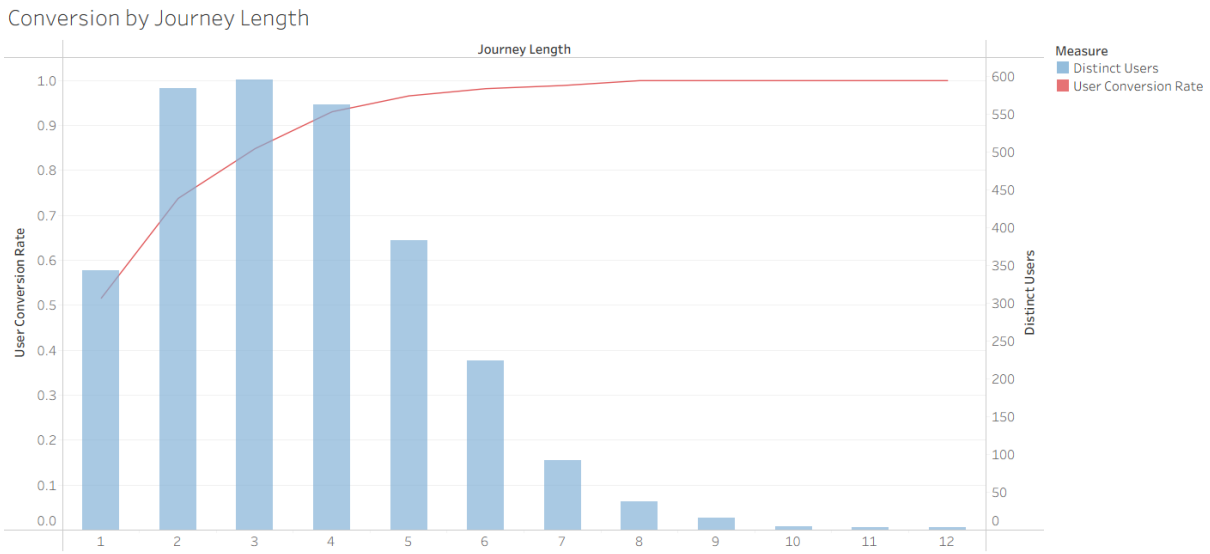


4.4.Journey length and conversion relationship

The journey-length chart shows that conversion rate increases as journey length increases. Users with more touchpoints have higher observed conversion.

At the same time, the number of users reduces as journey length grows, meaning most users have shorter journeys and long journeys are less frequent.

This supports a simple behavioural conclusion: conversion in this dataset is commonly associated with multiple interactions, so a single touchpoint position does not capture the full story.



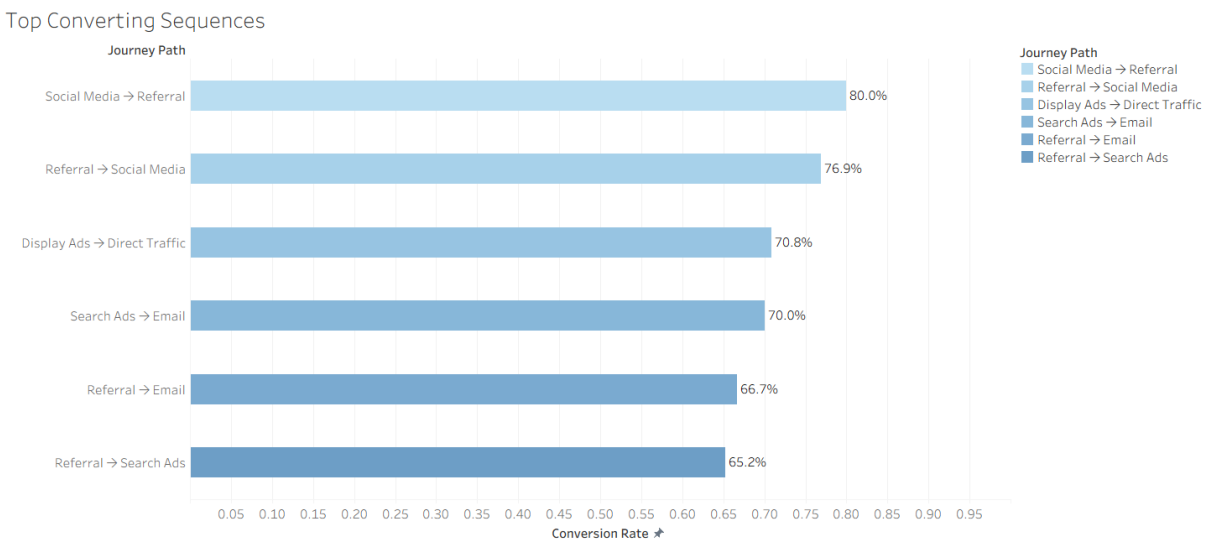
4.5.High-performing sequences (top paths)

The "Top Converting Sequences" view shows that certain two-step channel combinations have higher conversion rates than others, giving clearer separation than first-touch/last-touch.

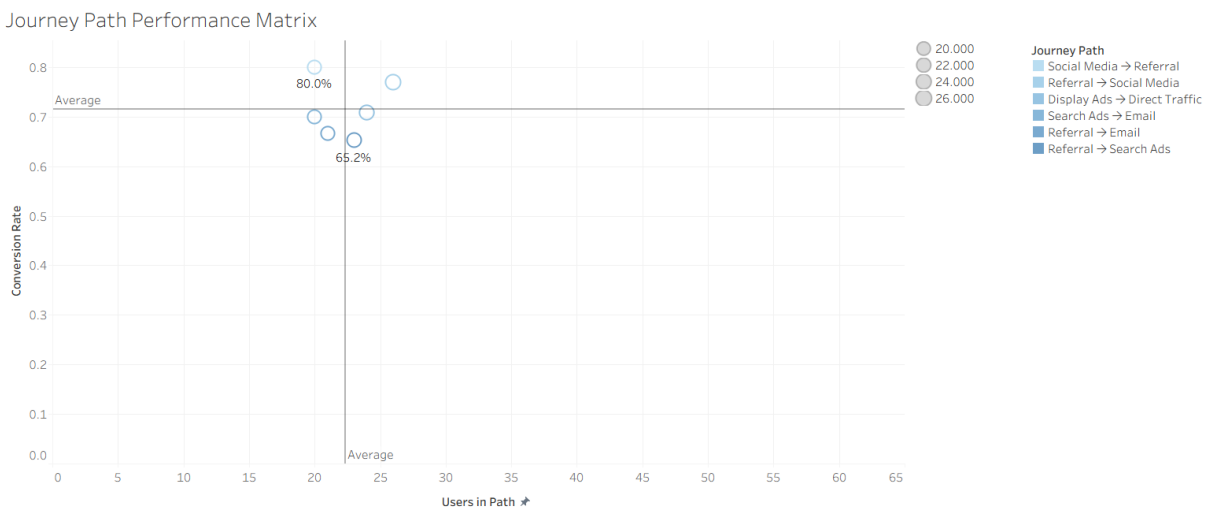
The strongest sequences shown include:

- Social Media → Referral (80.0%)
- Referral → Social Media (76.9%)
- Other paths like Display Ads → Direct Traffic (70.8%) and Search Ads → Email (70.0%) also perform well.

Since the dashboard applies a minimum user threshold (Users in Path ≥ 20), the top list focuses on paths that occur often enough to matter.



4.6.Path volume vs conversion matrix (prioritisation view)



The matrix compares each journey path using:

- Users in Path (how common the path is)
- Conversion Rate (how well the path converts).

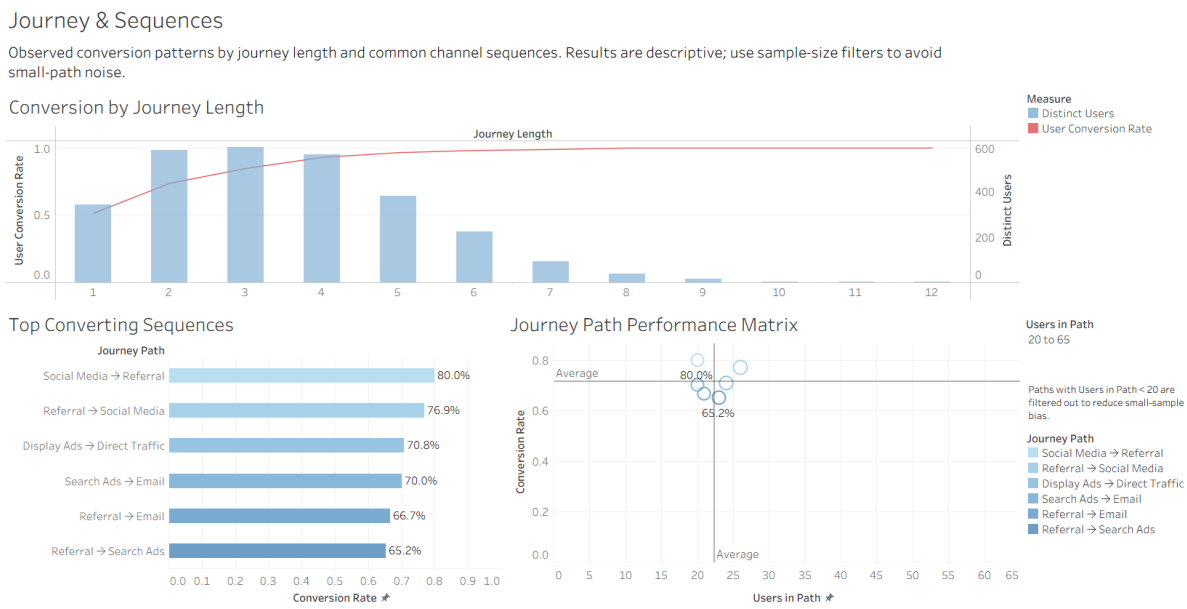
Average reference lines split the chart into four action zones:

1. High conversion + high volume: best candidates to prioritise and scale

2. High conversion + low volume: promising paths to expand and monitor
3. Low conversion + high volume: paths that need improvement or redesign
4. Low conversion + low volume: lowest priority

This view is useful because it balances performance with volume, so actions focus on what is both effective and impactful.

4.7.Overall conclusion from the visuals



The visuals show an overall user conversion rate of 83.6%. Because this baseline is high, the first-touch and last-touch views have only small differences between channels. In the first-touch view, Display Ads has 87.0% while Search Ads has 80.5%. Most other channels stay between 83% and 84%. The last-touch view follows the same pattern: Display Ads at 85.7% and Search Ads at 82.3%. This tells us that looking at a channel only at the start or only at the end of the journey does not separate performance well in this dataset. Single-touch rules therefore give limited insight.

Another pattern is that conversion rate rises as journey length increases. Users with more touchpoints show higher rates. At the same time, the number of users falls quickly after 3–4 touches. Most users stay in short to medium journeys. The practical point is that most conversions come from these shorter journeys because they contain more users.

The sequence-level views give more difference than the single-touch charts. The Top Converting Sequences chart shows bigger differences between paths. For example, Social Media → Referral has 80.0%, but Referral → Search Ads has only 65.2%.

The Journey Path Performance Matrix adds information. It places each path on two measures: number of users and conversion rate. This helps to find paths that have both many users and higher conversion rates, instead of choosing a path only because its rate is high from a small group of users.

Overall, the visuals give a shortlist of channel sequences and journey patterns that have higher conversion rates and enough user volume. These paths can be taken for further analysis and decision-making.

5. TABLEAU FEATURES AND JUSTIFICATIONS

5.1.Data sources

Two data sources are used in Tableau.

Data Source A: multi_touch_attribution_data.csv (touchpoint level)

This file keeps the data at event level (one row per touchpoint). It is needed for any view that uses time order, such as first-touch, last-touch and journey length. All events for each user are present, so user-level calculations can be done.

Data Source B: path_stats.csv (path level summary)

This file has one row for each complete journey path. It already contains Users in Path and Conversion Rate. This avoids the need to build long path strings or heavy calculations inside Tableau.

5.2.Calculated fields

Calculated fields are created so that all metrics are at user level. This stops users with many touchpoints from being counted more than once.

• Is Conversion: IF [Conversion] = "Yes" THEN 1 ELSE 0 END

Changes the text field to a number.

• User Converted: { FIXED [User ID] : MAX([Is Conversion]) }

Marks whether a user converted at least once in the journey.

• User Conversion Rate: COUNTD(IF [User Converted] = 1 THEN [User ID] END) / COUNTD([User ID])

Uses distinct users in numerator and denominator so the rate stays correct.

• First Touch Time and Is First Touch: selects the first event per user.

• Last Touch Time and Is Last Touch: selects the last event per user.

• Journey Length: counts the number of touchpoints per user.

• Has Arrow: checks for "→" to show only journeys with more than one step.

Note: If a user has two events at the exact same timestamp, first-touch and last-touch results can change. The charts are therefore shown as observed patterns in the dataset.

5.3.Filters, aggregations and sorting

• Campaign quick filter – lets users compare patterns by campaign in the same dashboard.

• Is First Touch / Is Last Touch filters – keep only one event per user in those charts.

• Users in Path ≥ 20 – removes very rare paths so conversion rates are more stable.

• Has Arrow = True – limits the Top Converting Sequences view to multi-step journeys.

Aggregations use COUNTD([User ID]) to count each user only once.

In path_stats.csv, MIN is used for Conversion Rate and Users in Path because the table is already at path level.

Sequences are sorted by conversion rate from high to low so the best paths appear at the top.

5.4.Interactivity and dashboard structure

Highlight actions are added. Clicking a channel or path highlights related marks so users can explore connections without losing the full view.

Labels and tooltips show the number of users (n) so readers can see whether the result is based on enough data.

Two dashboards are created:

• Channel Overview – baseline, first-touch and last-touch

• Journey & Sequences – journey length, top sequences and path matrix

This grouping lets business users compare related insights in one place using the same conversion definition.

6. DISCUSSIONS

6.1. Decision-making value

- Entry vs closing comparisons: The first-touch and last-touch views use the same user-level conversion metric. This lets stakeholders compare entry channels and closing channels without differences caused by the metric.
- Sequence prioritisation: The path table supports ranking and the matrix view. This helps marketing teams shortlist sequences that can be tested first, before making larger changes.
- Segmentation-ready reporting: The campaign filter allows users to review results by campaign. This supports decision discussions that are closer to real campaign planning and execution.

6.2. Limitations

- No incrementality or causality: This analysis is based on observed data. It cannot prove that a channel or sequence caused the conversion.
- Missing economic variables: The dataset does not include cost, impressions, or revenue. Because of this, the dashboard cannot calculate ROI metrics like CPA or ROAS. So, any budget recommendation should be treated as initial and not final.
- Filtering rare paths: Minimum sample size filters reduce instability in rates, but they can remove rare sequences. Some rare paths may still be important for specific user groups.

7. REFLECTION AND IMPROVEMENTS

Challenges

The first challenge was the dataset structure. The raw file is at touchpoint level, so one user can have many rows. If simple averages were used, results would be biased toward users with longer journeys. To address this, user-level logic (converted user flag and distinct user counts) was applied consistently across worksheets. This approach aligns with research emphasising that attribution should consider the full customer journey rather than isolated touchpoints (Anderl et al., 2016).

The second challenge was ordering. First-touch and last-touch analysis depends on correct timestamps. Missing or tied timestamps can result in more than one row qualifying as the first or last touch, which affects the charts. Because rule-based attribution methods can misrepresent channel contribution, especially when data assumptions are weak, the results were treated as descriptive rather than causal (Berman, 2018).

The third challenge was sequence analysis. Building full paths directly in Tableau from raw data can be complex and computationally slow, and the number of unique paths increases rapidly. To manage this, a path summary table (path_stats.csv) was created and rare paths were filtered using a minimum user threshold. This reduces small-sample noise and improves interpretability. Graph- and path-based modelling approaches highlight the importance of structured journey analysis rather than relying only on single-touch rules (Anderl et al., 2016).

Another limitation is the dataset scope. It covers a short time window and has a relatively high overall conversion rate, so channel-level differences appear small. Prior research shows that observational attribution data can mask true incremental impact when user intent and targeting differ across channels (Abhishek, Fader and Hosanagar, 2017). Therefore, strong business conclusions cannot be drawn from channel-only comparisons.

Potential Improvements

In future work, fields such as spend, impressions and revenue can be added to the dataset. These fields would allow calculation of ROI metrics like CPA and ROAS. The dashboards would then become more useful for business decisions.

Touchpoints that occur before conversion can be separated from those that occur after conversion. This would stop the mixing of interactions that may influence conversion with those that do not.

Time gaps between touchpoints (time between one touch and the next) can also be added. This may help explain the difference between short and long journeys. This follows the multi-stage funnel modelling approach (Abhishek, Fader and Hosanagar, 2017).

For stronger support to decisions, the top sequences should be tested with A/B testing or incrementality analysis. Research shows that patterns seen in the data do not always mean one channel caused the conversion (Berman, 2018).

If more data becomes available, advanced methods such as Markov-chain or Shapley-based attribution models can be built outside Tableau. The results can then be brought into dashboards for business use (Anderl et al., 2016).

8. CONCLUSION

This project studied multi-touch customer journeys to see how marketing channels and channel sequences are linked to conversion. The dataset was explored in Tableau. User-level conversion logic was used so that users with many touchpoints are not counted more than once.

The dashboards show that first-touch and last-touch views give only basic comparisons. They do not separate channel performance clearly in this dataset. Journey-level analysis gives more information. Users with more touchpoints show higher conversion rates. This supports the idea that conversion often happens after several interactions (Anderl et al., 2016). Sequence analysis finds specific paths with higher conversion rates. Low-volume paths were removed so that the results are more stable.

The main purpose of this project is to support business decisions. It helps users see how channels appear across the full journey and identify high-performing paths that can be tested further. The findings are descriptive only. The dataset does not support any claim that a channel caused the conversion. Any change in budget should first be tested with experiments or incrementality checks (Berman, 2018; Abhishek, Fader and Hosanagar, 2017).

This project achieved its goals. It provides dashboards and insights that show which marketing channels and touchpoint sequences are linked with higher conversion probability.

9. REFERENCES

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APPENDIX

- The source of dataset: Kaggle